

Exact optimal sparsity of real simplex tight frames via Haar matrices

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Status. *Literature-implied answer (complete real simplex tight-frame subfamily).* The exact formula below is an agent-identified translation of Cheon–Shader’s extremal theorem for sparse orthonormal matrices. It is not a claim of a new theorem. The translation settles the global optimal-sparsity problem for every real unit-norm tight frame of $n + 1$ vectors in $\mathbb{R}^n$.

1 Source problem and later algorithmic status

Casazza et al., *Optimally Sparse Frames* (arXiv:1009.3663), Section 5, printed pages 10–11 (PDF pages 10–11), asks whether and how Spectral Tetris extends when some eigenvalues are below 2. It notes that larger DFT blocks do not always give globally optimally sparse frames, even for tight frames, and asks for an extension appropriate “in the sense of optimal sparsity.” The two source pages are reproduced below.

The reader will have realized that Spectral Tetris generates frames which are ‘only’ optimally sparse with respect to the standard unit vector basis. This seems at first sight like a drawback. However, if sparsity with respect to a different orthonormal basis is required, Spectral Tetris can easily be modified to accommodate this request by using vectors of this orthonormal basis instead of the standard unit vector basis when filling in the frame vectors in Steps 5, 6 and 11 in STF (cf. Figure [1](#)). It is a straightforward exercise to show that this modified Spectral Tetris algorithm then generates a frame which is optimally sparse with respect to this new orthonormal basis.

4.4 Special Case: Constructing Optimally Sparse Tight Frames

In the special case of equal eigenvalues, i.e., of tight frames, with N elements in $\mathbb{R}^n$, all eigenvalues need to equal $\frac{N}{n}$ for the equality $\sum_{j=1}^n \lambda_j = N$ to be satisfied. The maximal block number can be easily computed to be $\gcd(N, n)$. Theorem [4.5](#) then takes the following form:

Corollary 4.6. *For $n, N > 0$, the frame $STF(N; \frac{N}{n}, \dots, \frac{N}{n})$ is optimally sparse in $\mathcal{F}(N, \{\frac{N}{n}\}_{i=1}^N)$ with respect to the standard unit vector basis. That is, this frame is $N + 2(n - \gcd(N, n))$ -sparse with respect to the standard unit vector basis.*

5 Conclusions and Discussion

In this paper we considered the design of frames which enable efficient computations of the associated frame measurements. This led to the introduction of the notion of a sparse frame as well as a sparsity measure for such frames, thereby introducing optimal sparsity as a new paradigm into the construction of frames. We then analyzed an extended version of Spectral Tetris for frames and proved that the frames constructed by this algorithm are indeed optimally sparse. This shows that Spectral Tetris can serve as an algorithm for computing frames with this desirable property, and our results prove that it is not possible to derive sparser frames through a different procedure.

We would finally like to point out that the analysis in this paper leads to several intriguing open problems for future research; a few examples are stated in the sequel.

- *Eigenvalues also smaller than 2.* It is still an open problem whether and how Spectral Tetris extends to sets of eigenvalues, if some eigenvalues are smaller than 2. The extension of the

Spectral Tetris algorithm by inserting larger DFT matrices than the previously exploited 2×2 -matrices, allowed for some partial results (see [11]). However, from the results in [11] it can be deduced that this procedure does not always lead to optimally sparse frames even in the case when all eigenvalues are equal, i.e., the tight frame case. Hence, extensive research will be necessary to introduce an appropriate – in the sense of optimal sparsity – extension of Spectral Tetris.

- *Extension to other classes of frames.* Depending on the application, other desiderata might be requested from a frame such as, for instance, equi-angularity. For such a class of frames, the question of an optimally sparse frame can and should similarly be posed.
- *Relative sparsity/Compressibility.* Taking numerical considerations and perturbations into account, it will be necessary to extend the notion of sparsity to relative sparsity/compressibility for frames and analyze optimality for such.

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Casazza–Fickus–Heinecke–Wang–Zhou (arXiv:1108.4061) subsequently removed the spectral restriction. Their Theorem 3.12 constructs a unit-norm frame with any positive prescribed spectrum satisfying the trace condition. Their Section 3.2 explicitly warns, however, that the general construction does not in general produce the sparsest possible frame. Thus the construction part of the source question was answered, while global optimal sparsity remained a separate issue.

2 Exact simplex result

For a synthesis matrix F , write $\text{sp}(F)$ for its number of nonzero entries. Changing the ambient orthonormal basis left-multiplies F by an orthogonal matrix, so minimizing over all frames in the standard basis is equivalent to minimizing the source’s basis-relative sparsity.

Theorem 1 (Exact simplex sparsity). *Let $n \geq 1$, let $m = n + 1$, and put*

$$r = \lceil \log_2 m \rceil.$$

Among all real unit-norm tight frames of m vectors in $\mathbb{R}^n$, the minimum number of nonzero synthesis-matrix entries is

$$\sigma(m) = m(r + 1) - 2^r. \quad (1)$$

A balanced (generally non-dyadic) Haar basis of the zero-sum hyperplane $\mathbf{1}^\perp \subset \mathbb{R}^m$ attains the minimum.

Proof. Let $F \in \mathbb{R}^{(m-1) \times m}$ be a unit-norm tight-frame synthesis matrix and set

$$U = \sqrt{\frac{m-1}{m}} F.$$

Then $UU^\top = I_{m-1}$. The matrix $P = U^\top U$ is a rank $m - 1$ orthogonal projection whose diagonal entries are all $(m - 1)/m$. Consequently $I - P = vv^\top$ for a unit vector v with $|v_j| = m^{-1/2}$ for every j . Choose a diagonal sign matrix D such that $D\mathbf{1}/\sqrt{m} = v$. The matrix

$$Q = \begin{bmatrix} UD \\ m^{-1/2} \mathbf{1}^\top \end{bmatrix}$$

is orthogonal, has a full constant last row, and satisfies $\text{sp}(Q) = \text{sp}(F) + m$.

Conversely, if an orthogonal $m \times m$ matrix has last row $m^{-1/2}\mathbf{1}^\top$ and first $m - 1$ rows U , then $F = \sqrt{m/(m-1)}U$ is a unit-norm tight-frame synthesis matrix: its rows are orthogonal with the required common square norm, and every column has norm one. Therefore the frame problem is exactly the orthogonal-matrix problem, with the m entries of the constant row removed.

Cheon–Shader’s main extremal theorem for orthogonal matrices with a full row (equivalently, after transposition, a full column) states that the minimum number of nonzero entries is

$$f(m) = (\lfloor \log_2 m \rfloor + 3)m - 2^{\lfloor \log_2 m \rfloor + 1} = m(r + 2) - 2^r. \quad (2)$$

Their extremizers are Haar-type matrices, and the bound is attained with the full row constant. Subtracting that row gives $\sigma(m) = f(m) - m$, which is (1).

For completeness, the matching construction can be described directly. Take a full binary tree with m leaves whose leaf depths differ by at most one. For every internal node with leaf set $S = A \dot{\cup} B$, use the unit vector supported on S which is constant on A , constant with opposite sign on B , and has mean zero. These $m - 1$ vectors form an orthonormal basis of $\mathbf{1}^\perp$. If $r = \lceil \log_2 m \rceil$, the tree has $2^r - m$ leaves at depth $r - 1$ and $2m - 2^r$ leaves at depth r . The total support of its internal-node vectors equals the sum of the leaf depths,

$$(2^r - m)(r - 1) + (2m - 2^r)r = m(r + 1) - 2^r.$$

Thus the construction realizes the lower bound with a constant full row. $\square$

3 Consequences for the low-redundancy question

The tight-frame eigenvalue is $m/(m - 1) < 2$ whenever $m \geq 3$, precisely the regime excluded by the source paper’s optimality theorem.

If one formally extrapolates the source’s high-redundancy tight-frame formula $N + 2(n - \gcd(N, n))$ to $N = m$, $n = m - 1$, it becomes $3m - 4$. The exact simplex formula agrees with it for $m = 2, 3, 4$ and is strictly larger for every $m \geq 5$. Indeed,

$$\sigma(m) - (3m - 4) = m(r - 2) - 2^r + 4 > 0 \quad (m \geq 5).$$

Hence the high-redundancy formula cannot extend verbatim below redundancy 2, even in the most rigid real tight-frame family.

For the first new case, $m = 5$ and $n = 4$, the exact optimum is 12, whereas the formal extrapolation is 11. An optimal orthonormal basis of $\mathbf{1}^\perp \subset \mathbb{R}^5$ consists of the four vectors supported, respectively, on

$$\{1, 2, 3, 4, 5\}, \quad \{1, 2\}, \quad \{3, 4, 5\}, \quad \{4, 5\},$$

with the balanced Haar coefficients. Scaling these rows by $\sqrt{5/4}$ gives the desired five-vector unit-norm tight frame in $\mathbb{R}^4$ with $5 + 2 + 3 + 2 = 12$ nonzero entries.

4 Provenance, limitations, and review focus

The source paper does not cite Cheon–Shader, and Cheon–Shader predates the frame question. The relation is the projection-completion identification in the proof above. The supporting authors did not know they were answering the source problem.

The exact result is limited to real simplex frames ($N = n + 1$). It does not settle optimal sparsity for arbitrary spectra below 2, for general low-redundancy tight frames, for complex frames, or for

the source’s other questions about equiangularity and compressibility. A bounded search covered the four run indexes, exact-title and formula searches, arXiv:1108.4061, and the later frame survey literature. No frame-theory source explicitly making this simplex/Haar identification was found, but the decisive extremal theorem has been known since 2000. The packet is therefore classified by literature provenance, not as a new partial solution.

Human review should check the equivalence between simplex frames and orthogonal completions with a constant row, and the use of the exact Cheon–Shader extremal formula. The latter is independently restated on page 213 of Cheon–Hwang–Rim–Shader–Song (2003), which attributes it to the 2000 Haar paper.

References

1. P. G. Casazza, A. Heinecke, F. Krahmer, and G. Kutyniok, *Optimally Sparse Frames*, IEEE Trans. Inform. Theory 57 (2011), 7279–7287; arXiv:1009.3663. See Section 5, printed pages 10–11.
2. G.-S. Cheon and B. L. Shader, *Sparse orthogonal matrices and the Haar wavelet*, Discrete Applied Mathematics 101 (2000), 63–76, doi:10.1016/S0166-218X(99)00182-1.
3. G.-S. Cheon, S.-G. Hwang, S.-H. Rim, B. L. Shader, and S.-Z. Song, *Sparse orthogonal matrices*, Linear Algebra Appl. 373 (2003), 211–222. See page 213 for the restatement of $f(m)$ and its attribution to the Haar paper.
4. P. G. Casazza, M. Fickus, A. Heinecke, Y. Wang, and Z. Zhou, *Spectral Tetris Fusion Frame Constructions*, J. Fourier Anal. Appl. 18 (2012), 828–851; arXiv:1108.4061. See Theorems 3.9 and 3.12 and Section 3.2.